Singh Priyanshu Kripashankar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.30
SDSM jr college or arts ,commerce and science	Class XII (maharashtra state board ,HSC)	2023	81.67%
Holy spirit english high school	Class X (maharashtra state board,SSC)	2021	91.80%

POSITIONS OF RESPONSIBILITY

Core Member - Elektronica | MEDTECH Domain, IIT Hyderabad

- Worked on embedded systems, biosignal processing, and biomedical engineering projects.
- Applied machine learning techniques to biosignal/sensor data for classification and analysis tasks.
- Attended skill development sessions to strengthen technical proficiency in embedded and machine learning tools.

Core Member - Cepheid | Astronomy & Astrophysics Club, IIT Hyderabad

- Conducted telescope observation sessions and astronomy outreach activities for students.
- Organized and facilitated discussions on astrophysics, observational astronomy, and space science topics.
- Delivered lecture talks on various space science topics to club members and the wider student community.

Core Member - IR Cell | Outbound Domain, IIT Hyderabad

- Assisted students with international opportunities and coordinated outreach, events, and application support.
- Co-organized Japan Day, managing logistics and coordinating a large group of participants and volunteers.

PROJECTS

Brain Tumor Detector (CNN)

- Built a CNN classifier in TensorFlow/Keras achieving 89% test accuracy and 0.88 F1 score on MRI images
- Resolved data scarcity and class imbalance via augmentation, scaling the dataset from 253 to 2,065 images
- Designed a lightweight custom architecture that outperformed ResNet50/VGG-16 transfer learning under CPU constraints

EMG Gesture Recognition for Unity Game Control

- Achieved 92-96% gesture classification accuracy by engineering 12 EMG features and training HistGradientBoosting classifiers on real-time 1000Hz signal data
- Cut control latency to under 100ms by building a multi-threaded pipeline with UDP-based Python-Unity communication
- Detected muscle fatigue with 85-90% accuracy using K-Means clustering on processed EMG signals

Globular Cluster Star Detection and Photometric Analysis

- Built an automated pipeline to detect stellar sources in Hubble imagery of the M12 cluster using Astropy and Photutils' DAOStarFinder
- Improved detection accuracy by reducing background noise via sigma-clipping statistical estimation
- Produced photometric brightness profiles and visual overlays of detected stars for flux analysis

SKILLS

Programming: Python, C/C++, MATLAB, SQL, Assembly

Tools & Technologies: Arduino, ESP32, Git,Github, Linux, Arduino IDE, LaTeX, EViews

Data Science & AI: NumPy, Pandas, Matplotlib, Scikit-learn, Machine Learning, Deep Learning, Reinforcement Learning, Signal Processing, Feature Engineering

Computer Science: Data Structures & Algorithms, Computer Networks, Computer Architecture, Relational Databases

Domains: Embedded Systems, Biomedical Signal Processing, Robotics, Computer Vision, Astronomy Data Analysis

Professional: Technical Communication, Event Coordination, Outreach, Stakeholder Engagement

RELEVANT COURSES

Computer Science & Artificial Intelligence: Introduction to Programming, Data Structures & Applications, Algorithms and Data Structure Lab, Artificial Intelligence, Computer Architecture *(Ongoing)*

Mathematics: Calculus I, Calculus II, Differential Equations, Elementary Linear Algebra, Transform Techniques, Introduction to Probability, Introduction to Statistics, Complex Variables, Mathematical Modeling & Systems Biology

Electrical & Embedded Systems: Introduction to Embedded Systems, Control Systems, Analog & Integrated Circuits, Linear Systems & Signals

Quantitative Finance & Analytics: Basic Econometrics and Forecasting

SCHOLASTIC ACHIEVEMENTS

- Active member of Bharat Scouts & Guides; participated in leadership, community service, outdoor training, and team-building activities *(2019-2020)*.

- Awarded A Grade in the Elementary Drawing and Geometry Grade Examination conducted by the Directorate of Art, Maharashtra *(2021)*

EXTRACURRICULAR

- Badminton - regional cluster tournaments and inter school events
- Football - regional cluster tournaments and inter school events